

Sporadic Abnormalities

Sporadic abnormalities are those that occur in singles, sometimes in pairs or in abundance, but they aren't repetitive (i.e., do not reappear in quick succession or after fixed intervals).

Sporadic abnormalities indicate underlying cortical dysfunction, though an associated structural lesion may not be always evident on neuroimaging (it should be sought) [1].

Three principal points of information are to be gained from the identification of sporadic abnormalities. These are as follows:

1. Where is the problem (i.e., location of the dysfunction)?
2. What is the problem (i.e., hint of potential etiology)?
3. Is there a predilection for epileptic seizures (i.e., epileptogenicity)?

The EEG cannot localize a lesion with the precision of modern neuroimaging techniques and electrographic abnormalities are notoriously nonspecific to etiology (only suggestive at best). The key role of the reader is to determine if an abnormality suggests a predilection for experiencing or developing epileptic seizures (epileptogenicity) as this determines the need for antiepileptic therapy [1].

"Sharpness" of the waveform at its peak (shape) is the main electrographic feature based on which this determination is made, the other being immediate surrounding activity.

Specifically look at the peak of the predominant phase of the discharge (the phase with the greatest amplitude), then measure its duration at the baseline. If the duration of the phase is less than 200 ms at baseline, categorize it as a *potentially* epileptiform abnormality (spike or sharp wave). The rest are slow waves. Slow waves with sharp peaks may be called "sharply contoured" but these do not qualify as spike or sharp waves [2].

Therefore, based on the sharpness of their peak and duration at baseline, sporadic abnormalities may be classified into two types:

1. spikes and sharp waves
2. slow waves

Figure 14.1 shows the key differences between spikes, sharp waves, and slow waves.

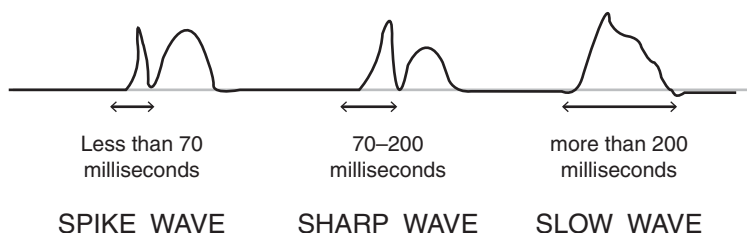


Figure 14.1 Types of sporadic abnormalities.

Spike and Sharp Waves

Spike and sharp waves have pointed peaks and are both epileptiform abnormalities or epileptic discharges. These are waveforms (or complexes, i.e., combination of waves), associated with a predisposition to experiencing epileptic seizures. It is important to understand that the mere presence of a spike or sharp wave is not pathognomic for epilepsy, it only means that there is a strong likelihood of experiencing an epileptic seizure. Conversely, a very small percentage of healthy people will have spike or sharp waves on their EEG [3].

Conventionally, spikes last more than 20 ms and less than 70 ms, while sharp waves last more than 70 ms but less than 200 ms (Figure 14.1). These are often followed by slow waves and together form a spike/sharp and slow-wave complex (called spike waves or sharp waves for short).

Sometimes there may be more than one spike in a complex, these are called polyspikes (Figure 14.2).

The terms spike and sharp waves may be used interchangeably and essentially mean the same thing – a predisposition to epileptic seizures. Often, both occur in the same disorder or the same patient.

Based on sharpness alone, many different waveforms including benign variants and artifacts may also resemble spike or sharp waves. When you suspect a spike or sharp wave, the next step is to confirm it as an epileptic discharge based on a few key electrographic features. We described these features in Chapter 13 for sharp variants. Let's revisit them again in the context of a spike or sharp wave.

Three key features on an epileptic discharge are as follows:

1. *Disruption of background activities:* Look at the immediate surrounding activity, before and after the discharge, epileptic discharges not only have a higher amplitude compared to the surrounding background activity but appear to disrupt it.
2. *Aftercoming slow wave:* Often, epileptic discharges will be followed by a negative slow wave of 150–350 ms duration and variable amplitude.

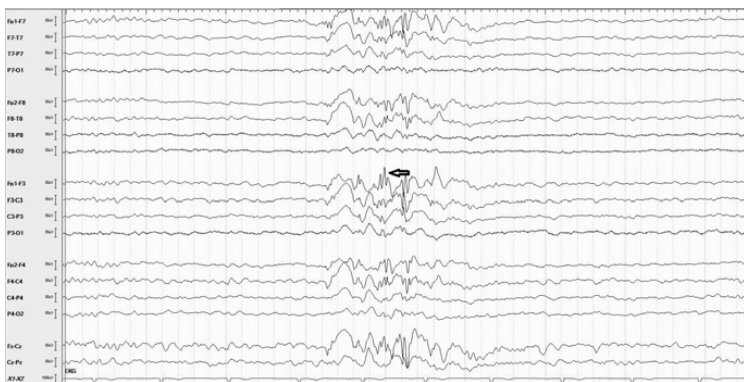


Figure 14.2 Twenty-three-year-old man with JME, frontal-predominant generalized polyspikes.

This may represent prolonged hyperpolarization due to secondary inhibition of a large neuronal population.

3. *Physiological field*: Epileptic discharges, as other forms of cerebral activity have physiological fields, this helps differentiate them from artifacts.

Other Features

- *Asymmetrical slopes*: The main component of the discharge has asymmetrical slopes with the first deflection being steepest.
- *Negative polarity*: The main component of the discharge is pointing upward. Rarely, positive spikes occur where the main component is positive (pointing downward); these may be seen after brain surgery, in neonates with periventricular hemorrhage or leukomalacia and children with multifocal epilepsies.
- Drops below the baseline.
- *Polyphasic*: crosses the baseline multiple times.
- Enhanced by drowsiness and non-rapid eye movement sleep [4].

Sharp waves that do not meet the criteria for a discharge may be called “sharp transients.” Figure 14.3 demonstrates the typical features of an epileptic discharge and Figure 14.4 shows positive spikes.

Once confirmed to be epileptic discharges, they can be further categorized based on their location:

1. focal
2. generalized

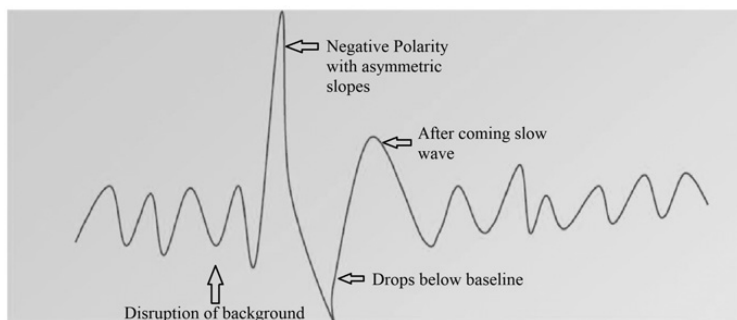


Figure 14.3 Features of an epileptic discharge.

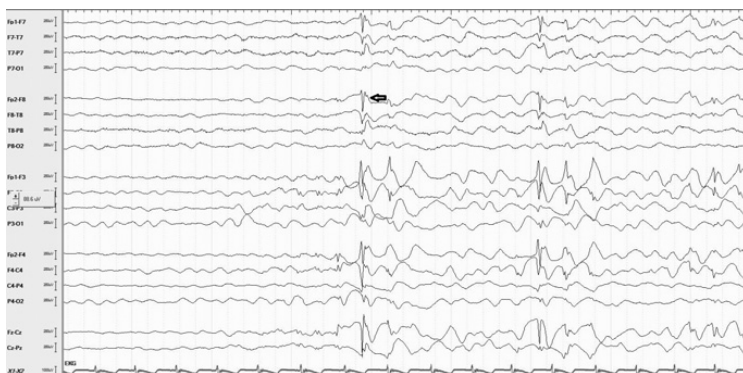


Figure 14.4 Seven-year-old boy with epilepsy, positive (downward) spikes.

Focal Discharges

Focal discharges originate from a single hemisphere and usually occur over a specific region. They may indicate focal epilepsy. If these occur independently over multiple regions, they are called multifocal discharges. The specific location of the focal discharge determines the strength of the association with focal epilepsy.

The strength of association with epilepsy is greatest for temporal and frontal discharges compared to other locations such as the centrottemporal (Rolandic), central, parietal, and occipital locations. For example, about 90% of those with anterior temporal spikes will have epilepsy compared to about

38% of those with Rolandic spikes. Approximately 60% of children with occipital spikes do not have epilepsy [5]. Occipital spikes are also common in nonepileptic conditions such as migraines and congenital blindness [6,7]. Approximately 94% of those with multifocal spikes have epileptic seizures [8]. Figure 14.5 shows a focal sharp wave (left temporal), and Figure 14.6 shows a focal spike wave (right temporal).

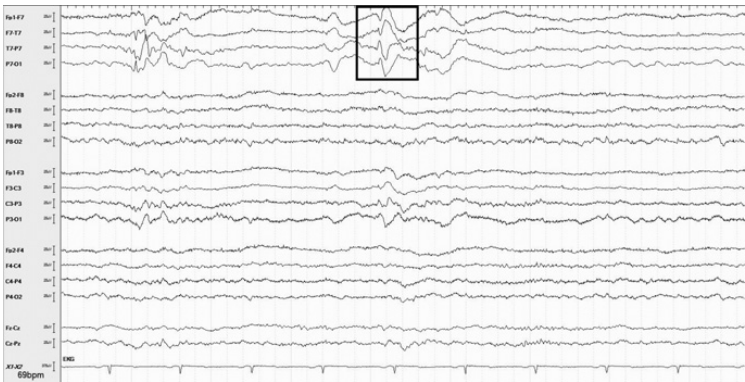


Figure 14.5 Fifty-year-old man with focal onset epilepsy, left temporal (T7) sharps.



Figure 14.6 Sixty-seven-year-old woman with a stroke, right temporal (T8) spikes.

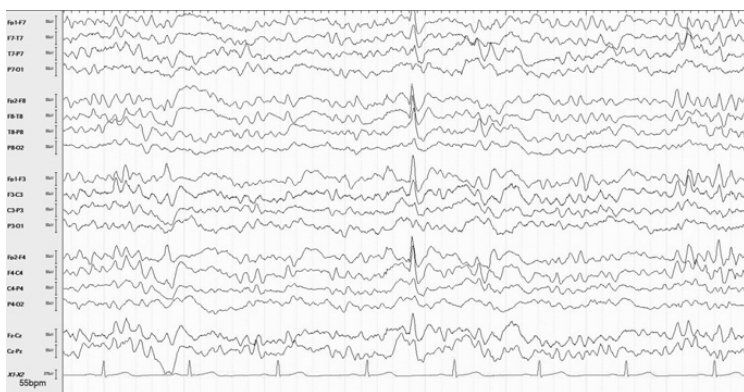


Figure 14.7 Thirty-one-year-old woman with LGS, frontal-predominant generalized sharps.

Generalized Discharges

These discharges originate simultaneously over both hemispheres, often maximal over the frontal regions and indicate generalized epilepsies. They may occur in singles, but more often in bursts of 1–3 s and can be longer. Longer runs (generalized spike waves) are typically associated with a subtle interference in mental function (absences). The frequency of the discharges determines the association with specific generalized epilepsies. Figure 14.7 shows a generalized sharp wave, Figure 14.8 shows a generalized spike wave.

Generalized spike and wave discharges (GSW) of approximately 2.5 to 3 Hz in frequency are the electrographic hallmark of typical absence seizures that occur in childhood absence epilepsy (Figure 14.9) [9].

Faster GSWs of approximately 4 to 6 Hz are seen with other generalized epilepsies such as juvenile myoclonic epilepsy (JME) and juvenile absence epilepsy (JAE), as shown in Figure 14.10 [10].

Slower GSWs of approximately 1.5 to 2.5 Hz are seen with Lennox–Gastaut syndrome (LGS) as shown in Figure 14.11 [11].

Generalized discharges may be activated or enhanced by photic stimulation and hyperventilation. Focal appearing or asymmetric fragments of generalized discharges are also common in those with generalized epilepsies [10].

Focal Discharges That Appear Generalized: In some acquired epilepsies such as posttraumatic epilepsy, focal discharges, particularly of frontal onset, can

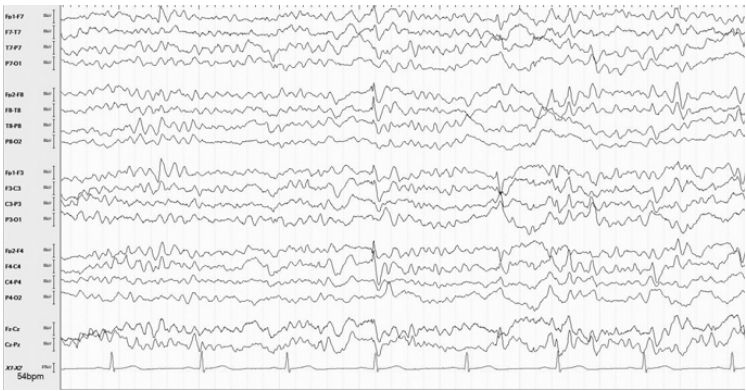


Figure 14.8 Same patient as in Figure 14.7, frontal-predominant generalized spikes.

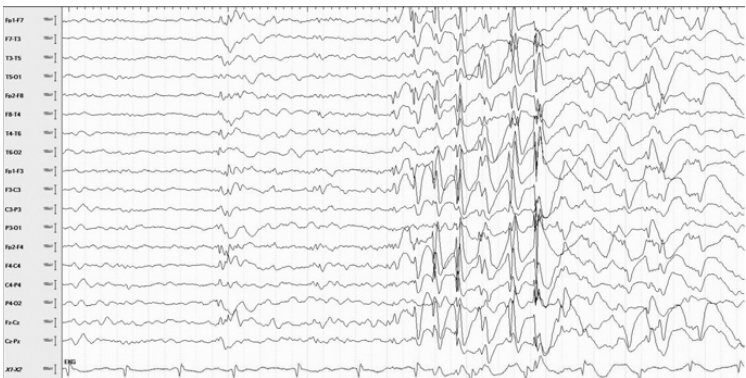


Figure 14.9 Eight-year-old boy with CAE, 3 Hz GSW discharges.

appear generalized. This happens when a focal or regional discharge directly leads to a corresponding contralateral discharge (bi-synchronous spike/sharp waves), this phenomenon is called as secondary bilateral synchrony. Another common situation occurs when a focal or regional discharge has a wide field that extends to the contralateral hemisphere, falsely appearing generalized.

Using a transverse montage, it is possible to distinguish such a wide field (large generator) from two discrete bi-synchronous generators. Often, other

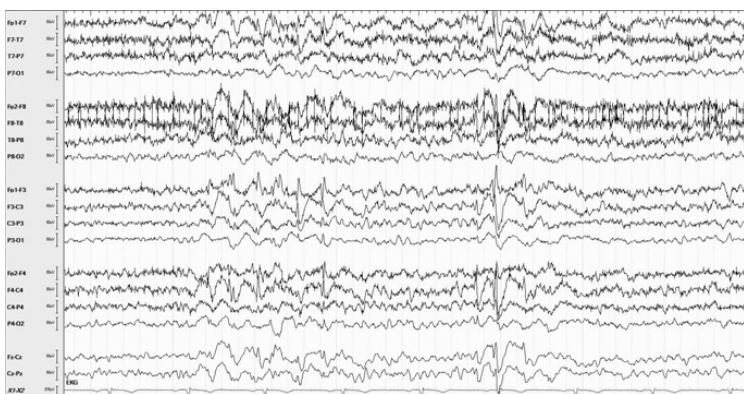


Figure 14.10 Seventeen-year-old boy with JME, 4 Hz GSW discharges.

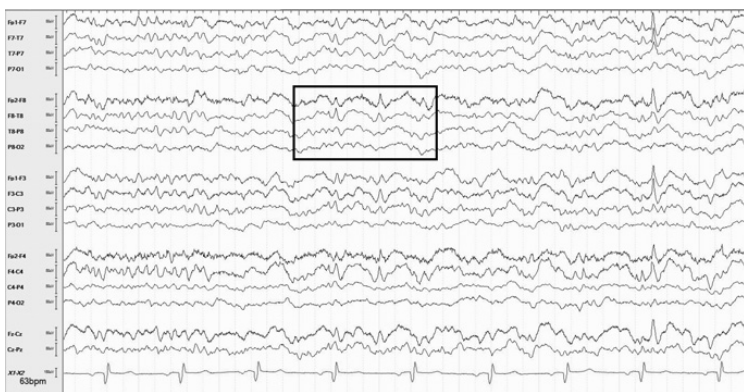


Figure 14.11 Same patient as in Figure 14.7, 2 Hz GSW discharges.

ipsilateral focal abnormalities such as slowing also suggest a focal rather than a true generalized discharge. Rarely, two distinct generators may occur in the same hemisphere to produce asynchronous focal discharges. Figure 14.12 demonstrates the difference between a generalized discharge, a focal discharge with a wide field and focal discharge with secondary bilateral synchrony. Figure 14.13 shows focal (right frontotemporal) discharges with wide contralateral fields that appear generalized.

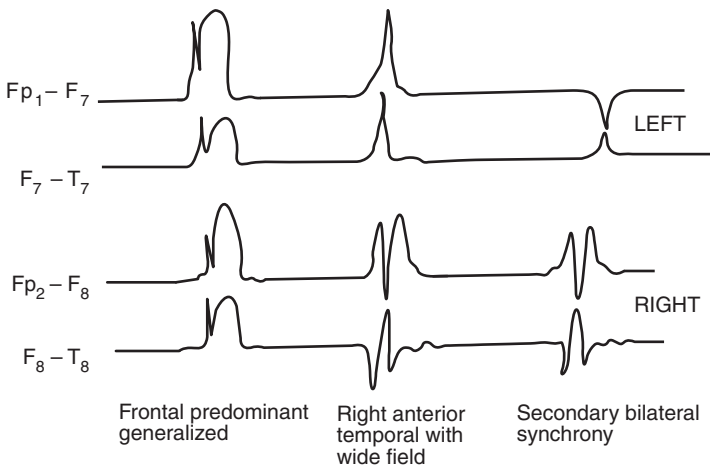


Figure 14.12 Types of bilateral appearing discharges.



Figure 14.13 Same patient as in Figure 14.6, right temporal discharges (T8) with wide fields.

Slow Waves

Slowing is the presence of abnormal delta or theta (slow) waves. Normal slow waves (physiological slowing) is seen in drowsiness or sleep and is described in Chapter 8.

“Delta slowing” was so named by W. G. Walter, its discoverer, because of its association with the three “Ds” of disease, degeneration, and death. Its presence indicates underlying cerebral dysfunction. Often multiple types of slowing coexist in a single record. Slowing by itself is typically not associated with epileptic seizures, though temporal-predominant lateralized rhythmic delta activity (previously called Temporal Intermittent Rhythmic Delta or TIRDA) is an important exception [12].

As with all foreground abnormalities, slowing should be described based on three key features:

1. location (focal, bi-hemispheric, generalized, or multifocal)
2. occurrence (sporadic or repetitive; see the following chapter)
3. morphology (polymorphic or monomorphic)

In addition, the constituent frequency of the slow waves (theta, delta, or mixed theta-delta waves) and reactivity to stimuli or drowsiness should also be noted. Further modifiers include the prevalence (continuous or intermittent), duration, amplitude, and frequency as described in Chapter 12.

Figure 14.14 shows intermittent left temporal (focal) polymorphic slowing, Figure 14.15 shows continuous left hemispheric (focal) polymorphic slowing and Figure 14.16 shows intermittent left temporal (focal) monomorphic slowing.

Bi-hemispheric (bilateral) slowing should be differentiated from generalized or diffuse slowing. Bi-hemispheric slowing, like generalized slowing appears in both hemispheres however it is typically not symmetric or synchronous. Generalized rhythmic slowing is further discussed in the following chapter (Chapter 15).

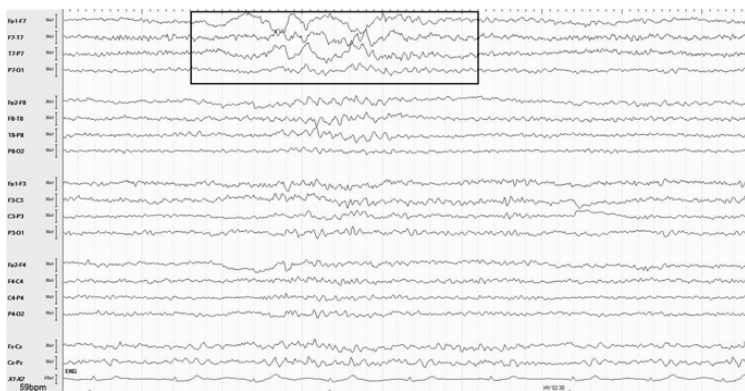


Figure 14.14 Sixty-two-year-old lady with a first-time seizure, intermittent focal left temporal polymorphic slowing.

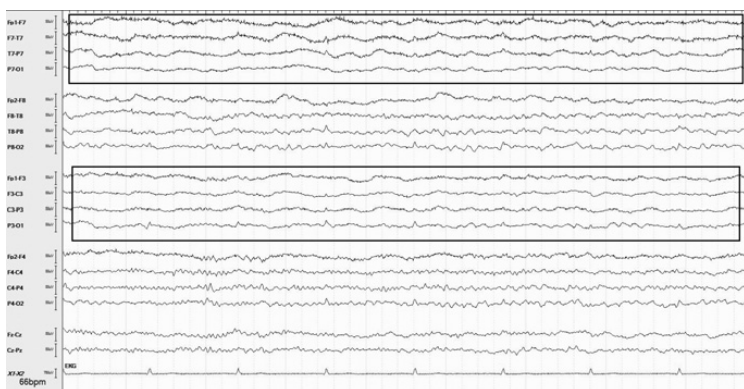


Figure 14.15 Sixty-eight-year-old man with a left hemispheric stroke, continuous left hemispheric polymorphic delta slowing.

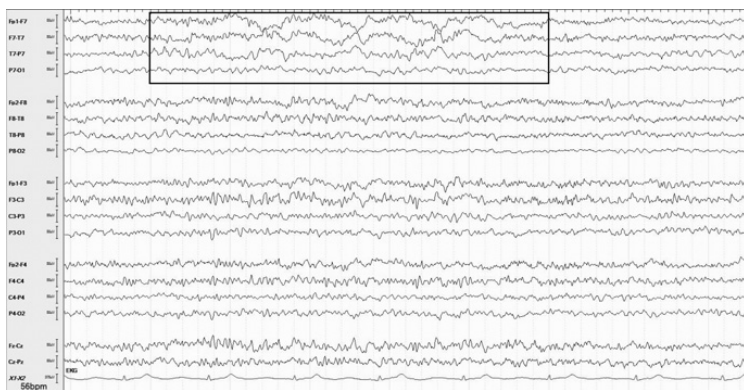


Figure 14.16 Same patient as in Figure 14.14, burst of left temporal monomorphic slowing.

Figure 14.17 shows intermittent bitemporal, but left>right polymorphic slowing, Figure 14.18 shows intermittent bursts of generalized slowing and Figure 14.19 shows continuous bi-hemispheric (bilateral) but left>right polymorphic slowing.

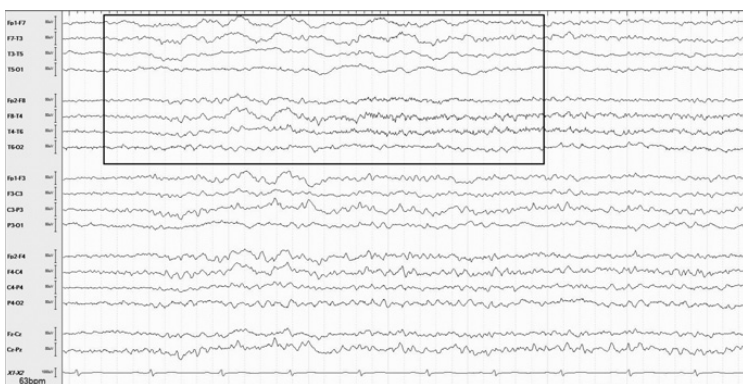


Figure 14.17 Eighty-two-year-old man with delirium, left greater than right bitemporal polymorphic delta slowing.

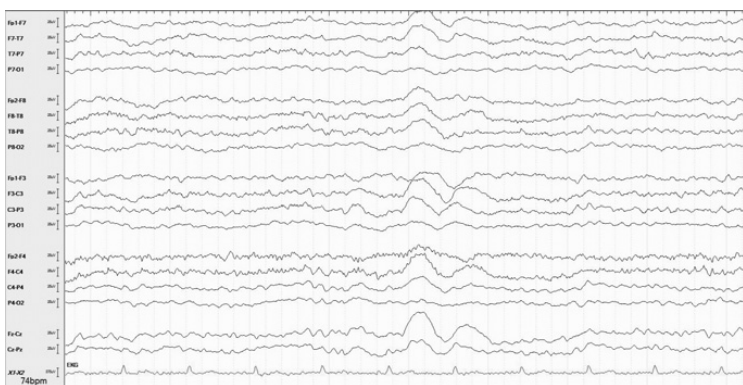


Figure 14.18 Same patient as in Figure 14.17, intermittent brief burst of generalized delta slowing.

Clinical Correlation

1. Generalized slowing diffusely affects both cerebral hemispheres, bi-hemispheric slowing is bilateral but affects each hemisphere in unequal measures, hence unlike generalized slowing, bi-hemispheric slowing is typically asynchronous and asymmetric.

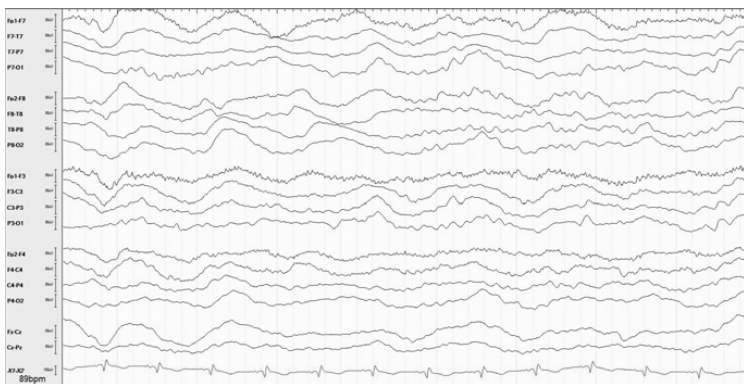


Figure 14.19 Seventy-year-old man with coma, continuous bi-hemispheric delta slowing.

2. Polymorphic slowing is a nonspecific sign of cerebral dysfunction, when focal, it suggests a localization.
3. Monomorphic (rhythmic) slowing may indicate epileptogenicity; more on this in the following chapter.
4. Continuous slowing may indicate static dysfunction (e.g., stroke or tumor), while intermittent slowing may be indicative of fluctuating dysfunction (e.g., postictal, migraines, transient hypoperfusion, or delirium).
5. Lower frequencies, higher amplitudes, and greater prevalence may correspond to the severity of dysfunction. A reversal of these features over time could be indicative of electrographic improvement.
6. Neither of the above features are sensitive or specific to the severity or etiology of the dysfunction.

Chapter Summary

1. Sporadic abnormalities are those that occur in singles, in pairs, or in abundance but are not repetitive.
2. They indicate cortical dysfunction, but the central question is epileptogenicity.
3. The key electrographic feature to classify the waveform is its sharpness.
4. Spike and sharp waves are pointed in morphology and are typically epileptogenic; these terms are often used interchangeably.
5. Slow waves are blunted in morphology and are typically not epileptogenic (except rhythmic focal slowing).

6. Spikes are 20 to 70 ms and sharp waves are 70–200 ms in duration; anything greater than 200 ms is a slow wave. Though spikes appear sharper/more pointed than sharp waves, they essentially mean the same thing – an association with epileptic seizures (not pathognomic for epilepsy).
7. Three key features of an epileptic discharge (sharp or spike wave) are its disruption of the normal background, an aftercoming slow wave and a physiological field.
8. Based on their location, epileptic discharges may be classified as focal (one hemisphere), generalized (both hemispheres), bi-hemispheric (both hemispheres but asynchronous), and multifocal.
9. The specific location of the focal discharge determines the strength of its association with focal epilepsy.
10. The frequency of the generalized discharge determines the specific association with generalized epilepsies.
11. Sometimes a focal discharge may appear generalized, as in secondary bilateral synchrony (two synchronous generators), or with a wide field (single large generator). A transverse montage helps distinguish these.
12. Sporadic slowing should be described based on its location (generalized, bi-hemispheric, focal, or multifocal) and morphology (monomorphic or polymorphic). It may be continuous or intermittent.
13. Polymorphic slowing is a nonspecific marker of cerebral dysfunction; monomorphic (rhythmic) slowing may indicate epileptogenicity.
14. Continuous slowing is indicative of static cortical dysfunction (e.g., stroke), while intermittent slowing may indicate fluctuating (dynamic) cortical dysfunction (e.g., postictal).
15. Electrographic features of slowing are neither sensitive nor specific to the severity/etiology of cerebral dysfunction. Slowing merely indicates the presence of cerebral dysfunction and in specific circumstances – epileptogenicity.

References

1. King MA, Newton MR, Jackson GD, et al. Epileptology of the first-seizure presentation: a clinical, electroencephalographic, and magnetic resonance imaging study of 300 consecutive patients. *The Lancet*. 1998 Sep 26;352(9133):1007–11.
2. Hirsch LJ, LaRoche SM, Gaspard N, et al. American clinical neurophysiology society's standardized critical care EEG terminology: 2012 version. *Journal of Clinical Neurophysiology*. 2013 Feb 1;30(1):1–27.
3. Sam MC, So EL. Significance of epileptiform discharges in patients without epilepsy in the community. *Epilepsia*. 2001 Oct 29;42(10):1273–8.
4. Niedermeyer E. Abnormal EEG patterns: epileptic and paroxysmal. In *Electroencephalography: basic principles, clinical applications, and related fields* (p. 255). Oxford University Press, Oxford; 2005.

5. Fois A, Malandrini F, Tomaccini D. Clinical findings in children with occipital paroxysmal discharges. *Epilepsia*. 1988 Oct;**29**(5):620–3.
6. Slatter KH. Some clinical and EEG findings in patients with migraine. *Brain*. 1968 Mar 1;**91**(1):85–98.
7. Kellaway P, Bloxson A, MacGregor M. Occipital spike foci associated with retrolental fibroplasia and other forms of retinal loss in children. *Electroencephalography and Clinical Neurophysiology*. 1955 Jan 1;**7**(3):469–70.
8. Pedley TA. Interictal epileptiform discharges: discriminating characteristics and clinical correlations. *American Journal of EEG Technology*. 1980 Sep 1;**20**(3):101–19.
9. Hedström A, Olsson I. Epidemiology of absence epilepsy: EEG findings and their predictive value. *Pediatric Neurology*. 1991 Mar 1;**7**(2):100–4.
10. Seneviratne U, Hepworth G, Cook M, D'Souza W. Atypical EEG abnormalities in genetic generalized epilepsies. *Clinical Neurophysiology*. 2016 Jan 1;**127**(1):214–20.
11. Ferlazzo E, Nikaronova M, Italiano D, et al. Lennox–Gastaut syndrome in adulthood: clinical and EEG features. *Epilepsy Research*. 2010 May 1;**89**(2):271–7.
12. Bazil CW, Herman ST, Pedley TA. Focal electroencephalographic abnormalities. In *Current practice of clinical electroencephalography* (pp. 303–47). Lippincott Williams and Wilkins, New York; 2003.